

Project Review: Energy SVD on FC1

1. The Big Picture

Hey Sir, here is the review report for our latest experiment. In this phase, we applied the Energy SVD (ESVD) compression specifically to the first Fully Connected layer (FC1) of our CNN. The main goal was to tackle the network's huge memory footprint. Instead of just guessing a threshold, we wrote an Auto-Tuner to sweep through different τ (tau) values to mathematically find the best cutoff point where we save the most memory without breaking the model.

2. Auto-Tuning: Finding the Sweet Spot

Our auto-tuner tested values from 0.99 down to 0.70. We told the script to stop if the raw accuracy dropped by more than 1.0%.

Threshold (τ)	Raw Accuracy	Accuracy Drop
0.99	98.85%	0.01%
0.95	98.82%	0.04%
0.90	98.81%	0.05%
0.85	98.72%	0.14%
0.80	98.62%	0.24%
0.75	98.60%	0.26%
0.70 (Winner)	98.41%	0.45%

The Verdict: The script confidently selected $\tau = 0.70$. This means we essentially threw away 30% of the mathematical "energy" in the singular values, yet the model's accuracy only dropped by 0.45% before any healing. The network is incredibly robust!

3. Let's Do the Manual Math

To prove the SVD logic is working exactly as intended, let's manually calculate the parameter reduction for the FC1 layer based on our chosen Rank ($k = 13$).

The Original FC1 Layer

- **Input features:** 3136 (from the flattened 64 channels * 7 * 7)
- **Output features:** 128
- **Weights:** $3136 * 128 = 401,408$
- **Biases:** 128
- **Total Original Parameters:** $401,408 + 128 = 401,536$

The Compressed FC1 Layer (Rank k = 13)

Because the maximum possible rank was 128, getting it down to $k=13$ is a massive compression. The SVD splits the layer into two smaller pieces:

- **Linear A (3136 $\rightarrow$ 13):** $3136 * 13 = 40,768$ weights (No bias)
- **Linear B (13 $\rightarrow$ 128):** $13 * 128 = 1,664$ weights + 128 biases = 1,792
- **Total Compressed Parameters:** $40,768 + 1,792 = 42,560$

The Net Savings

- **Parameters Deleted:** $401,536 - 42,560 = 358,976$
- **Final Model Size Check:** Our original whole model had 421,642 parameters. If we subtract the 358,976 we just deleted from FC1, we get exactly **62,666** total parameters remaining. This perfectly matches the torchinfo output!

4. Hardware Efficiency Results

Since we only compressed the dense layer, we expected a huge drop in memory size but only a small drop in computing speed (FLOPs). The results lined up perfectly with theory:

Metric	Baseline	ESVD Compressed	Total Reduction
Total Parameters (Memory)	421,642	62,666	85.14% Saved
Compute (MACs)	4,278,922	3,919,946	8.39% Saved

5. Healing & The Regularization Effect

Here is the coolest part of the experiment. After shrinking the network, we ran a gentle 5-epoch healing phase (learning rate = 0.00001) to let the surviving 13 singular values adapt.

- **Baseline Accuracy:** 98.86%
- **Raw Compressed Accuracy:** 98.41%
- **Healed Accuracy:** 99.14%

Conclusion: We didn't just save memory; we made the model **+0.28% more accurate** than when we started. By keeping only the top 13 dimensions, we mathematically deleted the "noise"

and overfitted data from the weights. The model was forced to generalize better, proving that Energy SVD acts as an incredible natural regularizer.